

Numerical study of a quayside wave energy recovery system in the presence of a dike

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Abstract

This study presents a numerical investigation of the hydrodynamic behavior of a quayside wave energy recovery system composed of a rectangular floating buoy oscillating near a rigid vertical dike in intermediate water depth. The main objective is to determine the dimensionless hydrodynamic coefficients (added mass, radiation damping, and excitation force) within the framework of linear potential theory, while explicitly analyzing the influence of key geometric parameters such as the floater's draft, width, and proximity to the dike. The modeling is based on the Boundary Element Method (BEM), enabling accurate simulation of wave diffraction and radiation phenomena induced by the floater's oscillations. To ensure consistency with the assumptions of the model, the analysis is carried out in the low-frequency regime, which also establishes the range of validity of the results. Numerical results show good agreement with existing experimental and analytical data, validating the proposed approach. The study reveals that the proximity of the dike amplifies wave interferences, significantly increasing radiation damping and excitation forces, especially for intermediate wavelengths. Additionally, increasing the floater's draft reduces the hydrodynamic coefficients and shifts their peak values toward longer wavelengths. Conversely, a wider floater enhances added mass, damping, and excitation force due to increased fluid interaction. These findings highlight the critical role of geometric configurations and environmental parameters in optimizing the efficiency of wave energy converters integrated into coastal infrastructures, aiming to enhance both energy harvesting performance and shoreline protection.

Keywords: Wave energy, Quayside system, Hydrodynamic coefficients, Boundary Element Method (BEM)

1. Introduction

Wave energy is considered one of the most promising sources of renewable energy due to its immense global potential. In order to collect and exploit this energy, several wave energy recovery devices have been developed (Falcão, 2010; Aderinto and Li, 2018; Zhao et al., 2019), among which quayside systems are attracting increasing interest. Although the energy potential of waves is generally higher in deep ocean environments, coastal devices, such as quayside systems, offer significant advantages. Their simplicity of installation, maintenance and connection contributes to a significant reduction in costs (Tournant et al., 2023), which explains their growing adoption. Furthermore, these systems help to protect coastal areas by attenuating wave energy before it impacts infrastructure, thereby limiting submersion, erosion and overtopping (Folley and Whittaker, 2009; Whittaker and Folley, 2012; Ozkan et al., 2020). Their integration into existing structures, such as dikes, can also improve their energy performance thanks to the partial reflection of waves (Elchahal et al., 2008; McIver and Evans, 1988). Additionally, wave reflections induced by the presence of the wall can increase the amplitude of the floater's motion in front of it, thus increasing the device's efficiency (Schay et al., 2013). The natural regularization of waves as they approach the shore promotes more stable and predictable energy production. Several recent studies have highlighted a growing focus on such solutions, driven by their cost-effectiveness, accessibility, and multifunctional nature. Moreover, to accurately assess the performance of a wave energy conversion system, it is important to take into account the interactions between the wave and the submerged or partially submerged

structures (Akarni et al., 2021, 2024), as these interactions have a direct influence on the overall performance of the system. The present study is part of this research framework.

Research on wave energy conversion systems is primarily focused around two main areas: energy studies and hydrodynamic studies. The first category aims to assess the energy potential of marine sites, optimize device performance, and evaluate the economic viability of wave energy converters. This includes analyzing wave characteristics, designing systems adapted to specific environmental conditions, and optimizing power take-off (PTO) mechanisms to maximize energy efficiency (Yu et al., 2024; Wang et al., 2025). On the other hand, hydrodynamic studies focus on understanding the interactions between waves and the structures of the converter (submerged or floating bodies, walls, damping plates, etc.), analyzing phenomena such as diffraction, radiation, resonance, and reflections, as well as evaluating hydrodynamic coefficients like added mass and radiation damping. These analyses are crucial for predicting the hydrodynamic loads experienced by the structures, adjusting design parameters, and ensuring the stability and durability of systems deployed at sea (Bhattacharjee and Soares, 2011; Koo and Kim, 2015).

Within these two research axes, numerous recent studies have significantly advanced knowledge. From a hydrodynamic perspective, Brown et al. (2017) investigated submerged heave plates used in wave energy converters (WECs), conducting experiments both in laboratories and at sea. Using the Morison equation, they determined the hydrodynamic coefficients (added mass and drag), highlighting strong intracycle variations, though constant values were found sufficient for accurately reconstructing experimental data. McCormick et al. (2018) proposed empirical expressions derived from the data of Vugts (1968), facilitating the analytical estimation of added mass and radiation damping for rectangular sections, in relation to strip theory. Neuvéglise et al. (2020) combined analytical modeling, ANSYS Fluent simulations, and experimental tests to analyze the behavior of an oscillating floater placed in front of a dike. Similarly, Chandrasekaran and Sricharan (2020) introduced a floating wave energy converter (FWEC) consisting of a central buoy and articulated floaters, exploring different floater shapes through a frequency and time model in WEC-Sim, showing the impact of hydrodynamic coefficients and the PTO system on device performance. Other recent studies have extended the energy analysis. Tournanta et al. (2021) performed numerical simulations of a quayside WEC using OpenFOAM, demonstrating that reducing the clearance distance improves performance by limiting overtopping. Masoomi et al. (2023) analyzed a hybrid system using OpenFOAM, demonstrating the influence of geometric configuration on system efficiency. Zhou et al. (2023) highlighted the power amplification generated by the interaction between a pitching cylindrical WEC and a parabolic breakwater. Meanwhile, Ulm et al. (2023) examined a new oscillating water column (OWC) geometry in a blue economy context, evaluating its performance through several indicators (extraction efficiency, amplification coefficient, etc.). Li et al. (2023) conducted combined tank tests and simulations for a double coaxial float WEC, analysing the effect of geometric parameters on energy absorption. Husain et al. (2023) studied the effects of variable geometry and PTO system characteristics on structural loads. Additionally, Mahdy et al. (2024) presented a comprehensive review of wave energy conversion devices, covering their modeling, associated projects, and global potential. Finally, Ji et al. (2024) compared two asymmetrical floating breakwaters integrating WECs, revealing that the L-shaped configuration performs better for short wave periods, while the trapezoidal shape optimizes the capture width for intermediate periods.

In continuation of the aforementioned research, the present study offers a numerical analysis of the hydrodynamic behavior of a wave energy conversion system installed along the quayside, in the presence of a rigid vertical dike. The system under investigation consists of a rectangular floating buoy, oscillating under the effect of incident waves in a fluid domain of intermediate depth. The aim of the analysis is to determine, in dimensionless form, the hydrodynamic coefficients of the system (the added mass, the radiation damping and the excitation force), using the Boundary Element Method (BEM). A notable contribution of this study lies in the detailed numerical formulation of the BEM approach, applied to a WEC system interacting with a rigid coastal structure. This rigorous methodological presentation, which is rarely detailed in previous studies, provides a better understanding of the influence of diffraction and radiation effects in a confined environment for low-frequency waves. The numerical results are compared with analytical and experimental data from the literature, showing good agreement and confirming the relevance and reliability of the proposed approach. This study thus contributes to the analysis and optimization of wave energy converters in coastal environments.

2. Theoretical formulation of the problem

We consider a rectangular floater of mass m , width $2l$, height h and draft d , located at a distance D from a

vertical dike. The floater only moves vertically, where the system is set in motion by the action of an incident harmonic wave of low amplitude a_i and wavelength λ . The water depth is noted as H . The study domain is delimited by a boundary $\Gamma = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4 \cup \Gamma_5 \cup \Gamma_6 \cup \Gamma_7$ and is defined in a Cartesian coordinate system (O, x, z) such that the axis (Oz) is directed vertically upward and the axis (Ox) is coincident with the position of the free surface at rest (see Fig. 1).

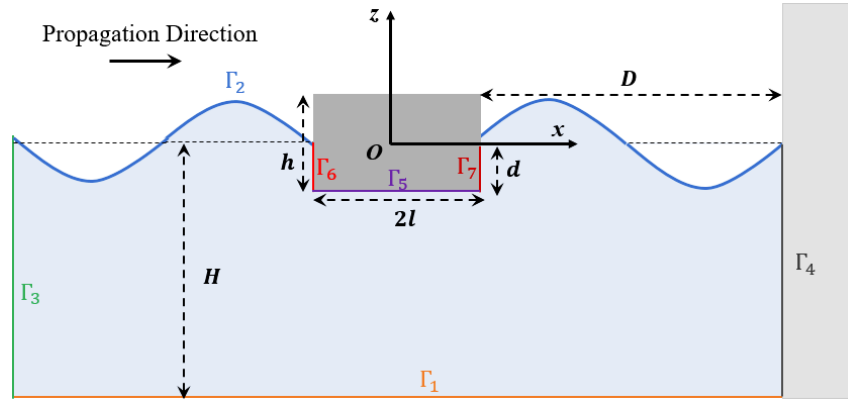


Fig. 1 Descriptive schema of the study domain.

The fluid is assumed to be perfect, homogeneous, inviscid, incompressible, and in a two-dimensional, irrotational flow. Under these assumptions, the flow field can be represented by a velocity potential such that:

$$\Phi(x, z, t) = R_e[\phi(x, z)e^{-i\omega t}] \quad (1)$$

Where R_e , i , ω and t are respectively the real part of a complex number, the complex number, the pulsation of the incident wave and the time.

In the case of a linear wave, the velocity potential is expressed according to the Haskind decomposition (Haskind, 1957; Newman, 1962) as follows:

$$\phi = \phi_I + \phi_D + \phi_R \quad (2)$$

The terms ϕ_I , ϕ_D and ϕ_R represent the incident wave, diffracted wave and radiated wave potentials respectively. The incident wave potential ϕ_I is defined by the following equation:

$$\phi_I = -i \frac{a_i g \cosh[k(z + H)]}{\omega \cosh(kH)} e^{ikx} \quad (3)$$

Where k is the wave number of the incident wave and it verifies the following dispersion relation:

$$\omega^2 = gk \tanh(kH) \quad (4)$$

The diffracted potential problem must satisfy the following equations (Zheng et al., 2004):

$$\left\{ \begin{array}{ll} \Delta \phi_D = 0 & \text{in } D \\ \frac{\partial \phi_D}{\partial n} = 0 & \text{on } \Gamma_1 \\ \frac{\partial \phi_D}{\partial n} = \frac{\omega^2}{g} \phi_D & \text{on } \Gamma_2 \\ \frac{\partial \phi_D}{\partial n} = ik \phi_D & \text{on } \Gamma_3 \\ \frac{\partial \phi_D}{\partial n} = -\frac{\partial \phi_I}{\partial x} & \text{on } \Gamma_4 \\ \frac{\partial \phi_D}{\partial n} = -\frac{\partial \phi_I}{\partial z} & \text{on } \Gamma_5 \\ \frac{\partial \phi_D}{\partial n} = \frac{\partial \phi_I}{\partial x} & \text{on } \Gamma_6 \\ \frac{\partial \phi_D}{\partial n} = -\frac{\partial \phi_I}{\partial x} & \text{on } \Gamma_7 \end{array} \right. \quad (5)$$

Where n is the external normal on the boundary that delimits the fluid domain Γ and g is the gravitational acceleration.

The set of boundary conditions for the diffracted potential problem can be expressed as follows:

$$\left\{ \frac{\partial \phi_D}{\partial n} \right\} = [F]\{\phi_D\} + \{S_D\}^T \quad (6)$$

Where: $\{S_D\} = (\{0\}, \{0\}, \{0\}, \{f(x, z)\}, \{f'(x, z)\}, \{f(x, z)\}, \{-f(x, z)\})$

With: $f(x, z) = -\frac{a_i g k \cosh[k(z+H)]e^{ikx}}{\omega \cosh(kH)}$ and $f'(x, z) = -\frac{ia_i g k \sinh[k(z+H)]e^{ikx}}{\omega \cosh(kH)}$

The wave radiation potential ϕ_R is written according to the following equation:

$$\phi_R = -i\omega a_R \varphi_R(x, z) \quad (7)$$

The radiated potential problem must satisfy the following equations (Zheng et al., 2004):

$$\left\{ \begin{array}{ll} \Delta \varphi_R = 0 & \text{in } D \\ \frac{\partial \varphi_R}{\partial n} = 0 & \text{on } \Gamma_1 \\ \frac{\partial \varphi_R}{\partial n} = \frac{\omega^2}{g} \varphi_R & \text{on } \Gamma_2 \\ \frac{\partial \varphi_R}{\partial n} = ik \varphi_R & \text{on } \Gamma_3 \\ \frac{\partial \varphi_R}{\partial n} = 0 & \text{on } \Gamma_4 \\ \frac{\partial \varphi_R}{\partial n} = 1 & \text{on } \Gamma_5 \\ \frac{\partial \varphi_R}{\partial n} = 0 & \text{on } \Gamma_6 \\ \frac{\partial \varphi_R}{\partial n} = 0 & \text{on } \Gamma_7 \end{array} \right. \quad (8)$$

The set of boundary conditions for the radiated potential problem can be expressed as follows:

$$\left\{ \frac{\partial \varphi_R}{\partial n} \right\} = [F]\{\varphi_R\} + \{S_R\}^T \quad (9)$$

Where: $\{S_R\} = (\{0\}, \{0\}, \{0\}, \{0\}, \{1\}, \{0\}, \{0\})$

It is observed that the set of boundary conditions for both the diffraction and radiation problems (systems 6 and 9) can be expressed as relationships between the potential and its normal derivative. This justifies the use of the Boundary

Element Method (BEM) in the present work.

2.1 Hydrodynamic coefficients

From Bernoulli's equation, we can deduce the hydrodynamic forces acting on the floater:

$$\begin{cases} F_{EX}(t) = -\rho \int_{\Gamma_5} \frac{\partial(\Phi_I + \Phi_D)}{\partial t} ds \\ F_R(t) = -\rho \int_{\Gamma_5} \frac{\partial \Phi_R}{\partial t} ds \end{cases} \quad (10)$$

Where F_{EX} , F_R and ρ are respectively the excitation force, the radiation force and the density of the fluid.

In linear theory, the floater position $z(t)$ is written in the form given by the equation below, where Z is the complex amplitude of the floater position.

$$z(t) = R_e[Z e^{-i\omega t}] \quad (11)$$

Similarly, the excitation force F_{EX} and the radiation force F_R can be written according to the equations below:

$$\begin{cases} F_{EX}(t) = R_e[f_{EX} e^{-i\omega t}] \\ F_R(t) = R_e[f_R e^{-i\omega t}] \end{cases} \quad (12)$$

The complex amplitude of the excitation force f_{EX} and the radiation force f_R are expressed below the floater as follows:

$$\begin{cases} f_{EX} = i\omega\rho \int_{\Gamma_5} (\phi_I + \phi_D) ds \\ f_R = i\omega\rho \int_{\Gamma_5} \phi_R ds \end{cases} \quad (13)$$

In the case of a monochromatic wave, the radiation force is also written as an added mass C_M and a damping C_D such that:

$$\begin{cases} C_M = \rho \int_{\Gamma_5} R_e[\varphi_R] ds \\ C_D = \rho\omega \int_{\Gamma_5} I_m[\varphi_R] ds \end{cases} \quad (14)$$

The added mass C_M , the damping coefficient C_D , and the excitation force f_{EX} are nondimensionalized according to the following expressions:

$$\begin{cases} C_m = \frac{C_M}{2\rho l d} \\ C_d = \frac{C_D}{2\rho\omega l d} \\ f_{ex} = \frac{|f_{EX}|}{2\rho g a l} \end{cases} \quad (15)$$

The complex amplitude of the floater position is written as:

$$Z = \frac{f_{EX}}{-\omega^2(C_M + m) - i\omega C_D + 2\rho gl} \quad (16)$$

With: $a_f = |Z|$ is the real amplitude of the floater's motion.

3. Numerical resolution

The fundamental principle of the Boundary Element Method (BEM) is to reformulate the partial differential equations governing the problem, originally defined over the entire fluid domain, into an integral equation expressed solely on the domain boundary. This transformation is achieved by applying Green's second identity, which allows partial differential equations, such as Laplace's equation in potential flow theory, to be rewritten in the form of boundary integral equations. As a result, there is no need to discretize the entire fluid domain, and the problem is reduced to solving for unknown quantities (such as the potential and its normal derivative) on the boundary.

The application of Green's second identity leads to the following integral expressions for ϕ_D and φ_R :

$$\begin{cases} c(p)\phi_D(p) = \int_{\Gamma} \left[\frac{\partial \phi_D(q)}{\partial n} G(r) - \frac{\partial G(r)}{\partial n} \phi_D(q) \right] ds \\ c(p)\varphi_R(p) = \int_{\Gamma} \left[\frac{\partial \varphi_R(q)}{\partial n} G(r) - \frac{\partial G(r)}{\partial n} \varphi_R(q) \right] ds \end{cases} \quad (17)$$

Here, $p(x, z)$ denotes the source point, while $q(x', z')$ represents a field point within the domain of study. The coefficient of the free term $c(p)$ is defined by:

$$c(p) = \begin{cases} 0 & p \equiv (x, z) \notin D \cup \Gamma \\ \frac{1}{2} & p \equiv (x, z) \in \Gamma \\ 1 & p \equiv (x, z) \in D \end{cases} \quad (18)$$

The expressions of the Green's function G and its normal derivative $\partial G / \partial n$, corresponding to the two-dimensional Laplace equation, are given by:

$$G = -\frac{1}{2\pi} \ln(r) \quad \text{and} \quad \frac{\partial G}{\partial n} = -\frac{1}{2\pi r} \frac{\partial r}{\partial n} \quad (19)$$

Where $r = \sqrt{(x - x')^2 + (z - z')^2}$ is the distance between p and q .

In order to numerically compute the velocity potential and its normal derivative along the boundary of the computational domain Γ , this boundary is discretized into a finite number of N segments of length Δl , with a singularity is imposed on each segment. The total number of segments is given by $N = \sum_{i=1}^8 N_i$ (see Fig. 2).

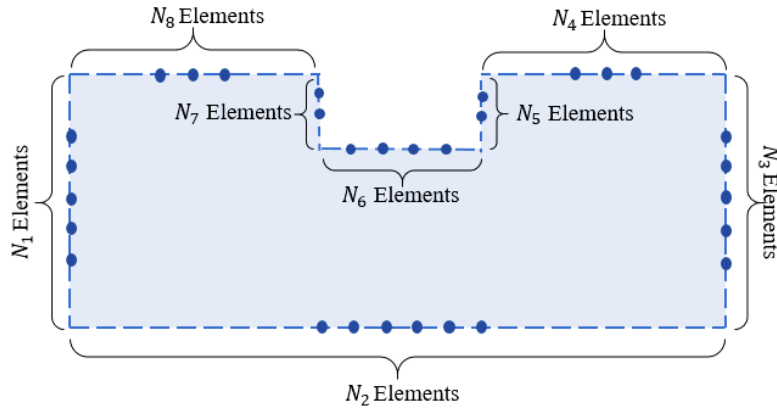


Fig. 2 Domain discretization.

There are several methods available for evaluating the boundary integrals on each segment. A common approach consists in placing a localized singularity at the center of each segment. This discretization reduces the boundary integral formulation to a linear system of equations, which is then solved to obtain the potential distribution along the boundary.

$$\begin{cases} c^i \phi_D^i = \sum_{j=1}^N E_{ij} \frac{\partial \phi_D^j}{\partial n_j} - H_{ij} \phi_D^j \\ c^i \varphi_R^i = \sum_{j=1}^N E_{ij} \frac{\partial \varphi_R^j}{\partial n_j} - H_{ij} \varphi_R^j \end{cases} \quad (20)$$

The explicit form of the influence matrices H and E is as follows:

- For $i \neq j$: $H_{ij} = \int_{(\Delta l_j)} \frac{\partial G}{\partial n} dl$ and $E_{ij} = \int_{(\Delta l_j)} G dl$
- For $i = j$: $H_{ij} = 2\pi$ and $E_{ij} = 2(1 - \ln \Delta l_j) \Delta l_j$

We can then transform Eq. (20) into a matrix form as follows:

$$\begin{cases} \{\phi_D\} = [K] \left\{ \frac{\partial \phi_D}{\partial n} \right\} \\ \{\varphi_R\} = [K] \left\{ \frac{\partial \varphi_R}{\partial n} \right\} \end{cases} \quad (21)$$

Where : $[K] = (\pi[I] + [H])^{-1}[E]$

With : $[I]$ is the identity matrix.

Each equation in this system consists of N equations with $2N$ unknowns, which are respectively the radiated potential and its normal derivative or the diffracted potential and its normal derivative. The additional N equations are obtained by writing the boundary conditions (Eq. 6 and Eq.9). Determining the diffracted and radiated potential therefore comes down to solving the following system:

$$\begin{cases} [L]\{\phi_D\} = \{S'_D\} \\ [L]\{\varphi_R\} = \{S'_R\} \end{cases} \quad (22)$$

Where : $[L] = ([F] - [K]^{-1})[K]$; $\{S'_R\} = -[K]\{S_R\}^T$ and $\{S'_D\} = -[K]\{S_D\}^T$

4. Results and discussion

In order to ensure the reliability of the numerical results obtained using the BEM, a convergence analysis was carried out to verify the independence of the results with respect to mesh refinement. Table 1 presents a grid independence study showing the variation of the floater's response a_f/a_i as a function of the total number of boundary elements N . The geometric parameters used are identical to those in Fig. 3.

The results indicate that the numerical value of a_f/a_i gradually converges toward the experimental value reported by Neuvéglise et al. (2020) for the dimensionless pulsation $\omega_1 = 0.6$. The relative error decreases significantly, from 5.15% for $N = 60$ to only 0.057% for $N = 1250$. Beyond $N = 600$, the variations in the results become negligible (relative error below 0.25%), demonstrating satisfactory mesh convergence. Accordingly, a mesh with $N = 600$ boundary elements was adopted for the remainder of the simulations, as it offers an optimal balance between computational efficiency and accuracy.

Table 1 Grid independence study.

Number of Elements N	60	90	120	240	520	600	900	1200	1250
Numerical value of a_f/a_i	2.1619	2.1905	2.2010	2.2454	2.2725	2.2741	2.2756	2.2762	2.2807
Relative error (%)	5.15	3.89	3.43	1.49	0.301	0.231	0.165	0.139	0.057

4.1 Validation

To validate the numerical approach proposed in this work, a comparative study was carried out with the experimental and analytical results reported by Neuvéglise et al. (2020). Figure 3 presents a comparison of the dimensionless amplitude of the floater's motion a_f/a_i as a function of the dimensionless wave pulsation $\omega_1 = \omega\sqrt{d/g}$, for a floater of width $2l = 0.1\text{ m}$ and draft $d = 0.25\text{ m}$, installed at a distance $D = 0.1\text{ m}$ from a vertical dike, in a water depth fixed at $H = 0.6\text{ m}$. Figure 4, on the other hand, shows a comparison of the dimensionless damping coefficient C_d as a function of the ratio H/λ , for a floater of width $2l = 3.334 H$ and draft $d = 0.4 H$, located at a distance $D = 0.2 H$ from a vertical dike.

These comparisons highlight a good agreement between the numerical results obtained using the BEM and the experimental and analytical data reported by Neuvéglise et al. (2020), particularly at low frequencies, as shown in Fig. 3. This confirms the effectiveness of the developed computational code in analyzing the dynamic behavior of the floater in the presence of a vertical dike within this valid range. However, discrepancies arise for frequencies above $\omega_1 > 0.6$, which can be attributed to several factors. which can be attributed to the absence of nonlinear effects (such as viscous damping, turbulence-induced losses, and partial wave breaking) that are not considered in the boundary conditions of the current linear potential flow formulation.

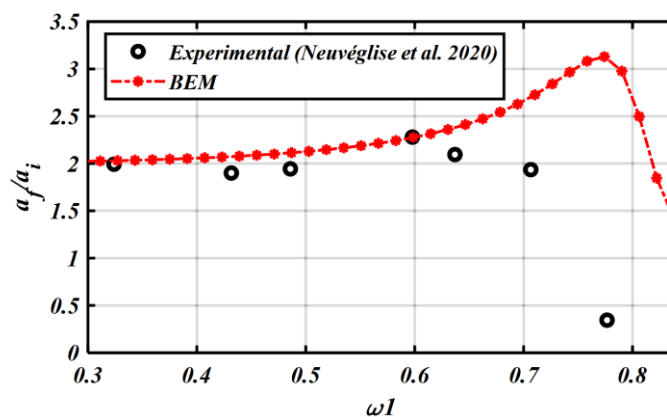


Fig. 3 Comparative study of the floater response a_f/a_i as a function of the dimensionless pulsation ω_1 .

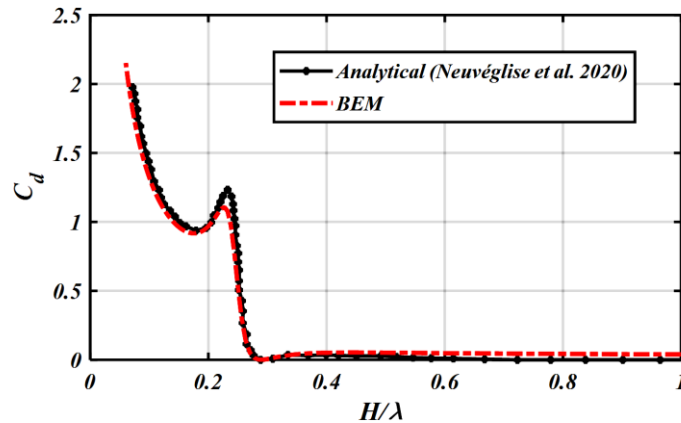


Fig. 4 Comparative study of the dimensionless damping coefficient C_d as a function of the ratio H/λ .

4.2 Hydrodynamic effect of the dike

In this part, the influence of the dike on the dimensionless hydrodynamic coefficients is analyzed. Simulations were carried out with the floater's draft fixed at $d = 0.4 H$ and its half-width at $l = 0.25 H$, while the distance D between the floater and the dike was successively set to $0.2 H$, $0.4 H$ and $0.6 H$.

Figure 5(a) shows that the distance D between the floater and the dike significantly influences the dimensionless added mass coefficient. For a reduced distance ($D/H = 0.2$), the coefficient exhibits a strongly pronounced and negative minimum around $H/\lambda \approx 0.27$, indicating significant interference effects between the incident, diffracted, and reflected waves. As D/H increases (0.4 and 0.6), this minimum becomes progressively less pronounced, with its absolute value decreasing by approximately 33%, and tends to smooth out, with a slight shift toward lower H/λ ratios. Moreover, for $H/\lambda > 0.5$, all curves converge toward positive values, indicating a weakening of the dike's effect on short waves. This behavior is explained by the interference generated between the incident waves, the waves reflected by the dike, and those diffracted around the floater. When the dike is close, these interactions are intense and significantly alter the pressure field around the floater, potentially causing a negative added mass effect. As the dike is moved farther away, this coupling weakens, and the system's behavior approaches that of an unbounded fluid domain.

Figure 5(b) shows that the dimensionless radiation damping coefficient generally decreases as the distance D between the dike and the floater increases. When the dike is close ($D/H = 0.2$), the damping is significantly higher, especially in the range $0.1 < H/\lambda < 0.3$, where a distinct peak is observed. As the dike moves farther away ($D/H = 0.4$ and 0.6), this peak becomes less pronounced and shifts slightly, while all curves tend to converge toward low and similar values for $H/\lambda > 0.35$, indicating a loss of the dike's influence for very short waves. This effect is due to the fact that a nearby dike enhances radiation effects by creating constructive interference between the waves emitted by the motion of the floater and those reflected by the dike. This leads to increased energy dissipation, reflected in a higher damping coefficient. As the dike moves away, this interaction decreases, resulting in reduced radiation dissipation. Thus, the proximity of the dike plays a critical role in determining the intensity of hydrodynamic damping, especially for intermediate to long waves.

Figure 5(c) illustrates the effect of the distance between the floater and the vertical dike on the dimensionless excitation force. When the dike is close to the floater ($D/H = 0.2$), the curve shows a pronounced initial peak reaching approximately 1.8 at $H/\lambda = 0.1$, followed by significant oscillations. As the distance D increases ($D/H = 0.4$ and $D/H = 0.6$), this initial peak decreases to around 1.75 and 1.6, respectively, and the oscillations become less intense. The excitation force then tends to smooth out, particularly for $H/\lambda > 0.3$, where the curves progressively converge. For instance, at $H/\lambda = 0.5$, the three configurations exhibit similar values around 0.45, indicating a reduced amplifying effect of the wall on short waves. This is due to the interference between the incident wave and the wave reflected by the dike. When the dike is near, these interferences are more intense and localized, leading to a local amplification of the excitation force. Conversely, a greater distance reduces this constructive interaction, thereby diminishing the amplifying effect of the dike.

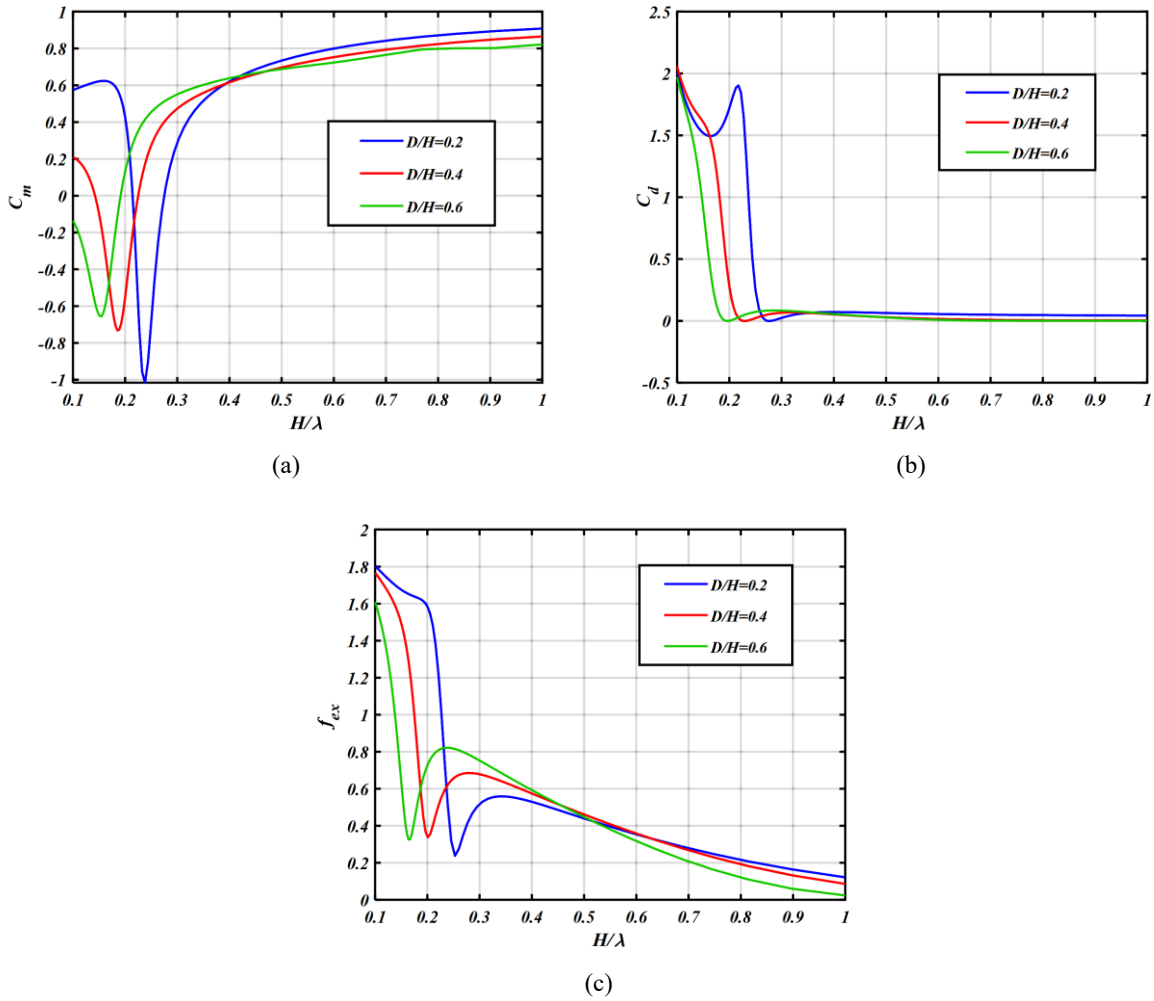


Fig. 5 Dimensionless hydrodynamic coefficients as a function of H/λ . (a): dimensionless added mass coefficient C_m (b): dimensionless damping coefficient C_d and (c): dimensionless excitation force f_{ex} .

4.3 Hydrodynamic effect of the draft

The hydrodynamic effect of floater's draft on the dimensionless hydrodynamic coefficient is illustrated in Fig. 6(a), Fig. 6(b), and Fig. 6(c). The simulations were carried out with a floater of half-width $l = 0.25 H$ and a floater–dike distance $D = 0.2 H$, while the floater draft d takes the values $0.2 H, 0.3 H$ and $0.4 H$, respectively.

Figure 6(a) shows that increasing the draft d of the floater has a significant effect on the dimensionless added mass coefficient C_m . Specifically, when the ratio d/H increases from 0.2 to 0.4, the absolute minimum value of C_m decreases markedly, from approximately 1.5 for $d/H = 0.2$ to around 1.1 for $d/H = 0.4$. This minimum also shifts toward lower values of H/λ : for $d/H = 0.2$, it is located around 0.31, while for $d/H = 0.4$, it reaches approximately 0.24. This behavior can be attributed to the increase in the submerged volume of the floater, which intensifies its interaction with the surrounding water column, modifying the distribution of hydrodynamic pressures on its surface. In addition, the lower attenuation of long waves at depth increases the sensitivity of the hydrodynamic response of the system to variations in draft, which explains the observed shift in the minimum of the added mass coefficient towards conditions associated with long wavelengths.

Figure 6(b) highlights the influence of the float's draft d on the dimensionless damping coefficient C_d . It is observed that increasing the ratio d/H from 0.2 to 0.4 leads to a significant reduction in the maximum value of C_d , which decreases from approximately 4 for $d/H = 0.2$ to 2.6 for $d/H = 0.3$, and further to 2 for $d/H = 0.4$, corresponding to an overall reduction of about 50%. Additionally, the damping peak shifts towards lower values of H/λ , moving from approximately 0.29 for $d/H = 0.2$ to around 0.21 for $d/H = 0.4$. This trend can be explained by the

effect of depth on the wave radiation mechanisms generated by the movement of the floater. A more submerged floater has a larger surface area interacting with the water column, which modifies the distribution of the radiated pressure field. This leads to energy dissipation by radiation that is more distributed and less intense locally, thus reducing the maximum value of the damping coefficient. Furthermore, in the case of long waves, whose influence penetrates more deeply, the interaction between the floater and the induced velocity field is more uniform, which contributes to the shift of the damping peak towards low H/λ ratios.

Lastly, it is observed that increasing the draft also reduces the maximum value of the dimensionless excitation force, decreasing from approximately 0.78 for $d/H = 0.2$ to around 0.58 for $d/H = 0.4$. Moreover, the peak of this force shifts towards lower values of H/λ , an effect that becomes particularly noticeable for $H/\lambda > 0.25$ (see Fig. 6(c)). This evolution can be explained by the vertical distribution of the dynamic pressure exerted by the wave field: a more submerged floater interacts with a larger portion of the water column, resulting in a more uniform distribution of wave forces. Short waves, whose energy is concentrated near the surface, become less effective at generating a significant excitation force on a deeply submerged floater, thus reducing the peak. In contrast, long waves, which reach deeper, maintain a strong interaction with the floater, which explains the shift toward lower H/λ ratios.

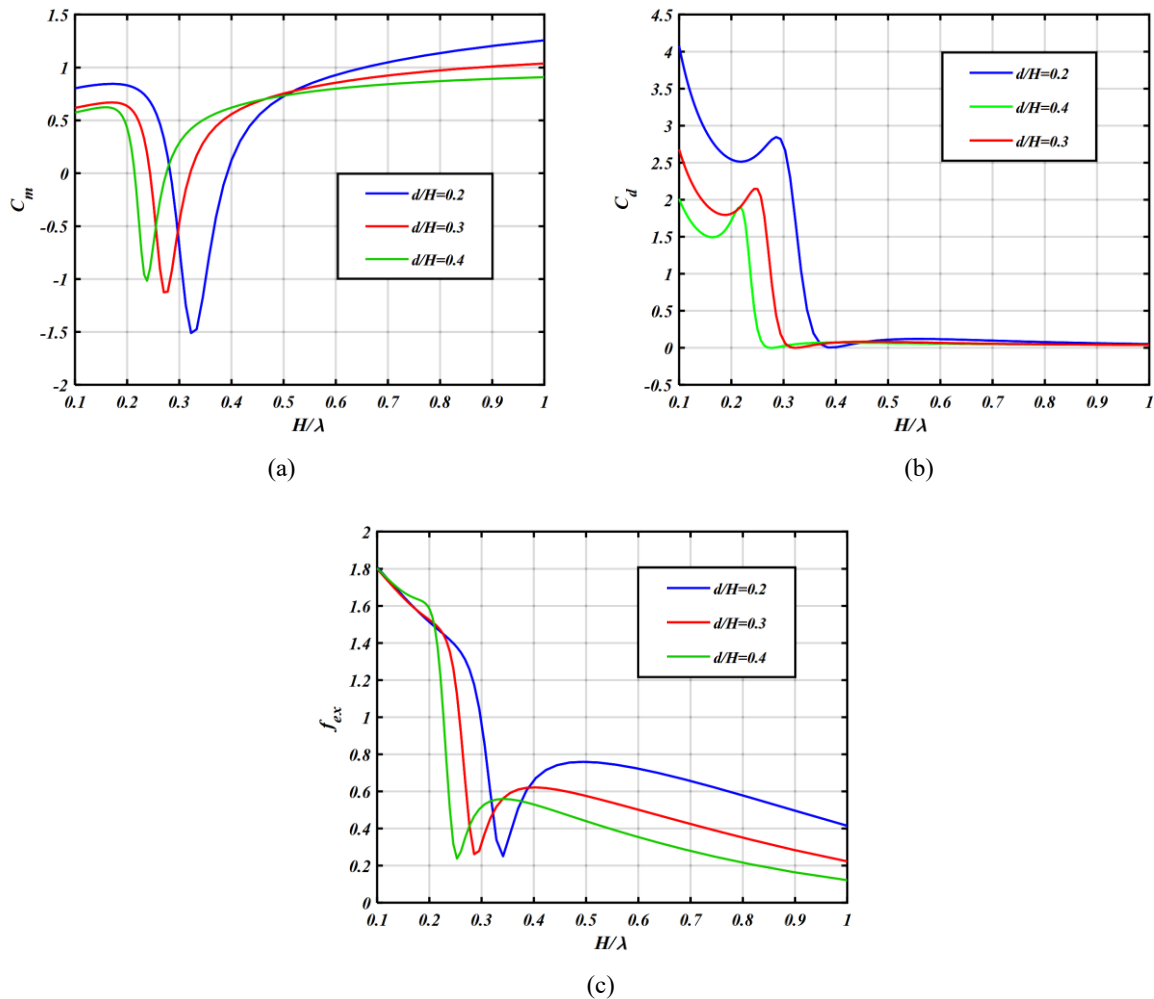


Fig. 6 Dimensionless hydrodynamic coefficients as a function of H/λ . (a): dimensionless added mass coefficient C_m (b): dimensionless damping coefficient C_d and (c): dimensionless excitation force f_{ex} .

4.4 Hydrodynamic effect of the floater width

The influence of the floater width on the dimensionless hydrodynamic coefficients is illustrated in Fig. 7(a), Fig. 7(b), and Fig. 7(c). Simulations were performed with a fixed floater draft of $d = 0.4 H$ and a constant distance from the dike $D = 0.2 H$, while the half-width of the floater l successively takes the values $0.125 H$, $0.25 H$ and $0.375 H$.

It is observed that increasing the floater width leads to a significant rise in the absolute value of the minimum dimensionless added mass coefficient, with this value increasing by nearly a factor of eight. Specifically, it increases from approximately 0.26 for $l/H = 0.125$ to around 2.1 for $l/H = 0.375$. This increase is accompanied by a slight shift of the minimum toward lower H/λ ratios, associated with longer wave conditions. This behavior can be attributed to the increased lateral surface area of the floater in contact with the fluid, which enhances the hydrodynamic inertia of the system. A wider floater displaces more fluid transversely during oscillations, resulting in a higher added mass. Furthermore, this enlarged volume interacts more effectively with the deeper pressure field components of long waves, explaining the observed shift toward lower H/λ values.

Additionally, increasing the width of the floater leads to a significant rise in the peak value of the dimensionless damping coefficient, by up to a factor of five. Specifically, this value increases from approximately 0.7 for $l/H = 0.125$ to around 3.6 for $l/H = 0.375$. A wider floater experiences stronger interactions with the incident waves, which amplifies the amount of energy radiated as secondary waves during its motion. This increase in radiated energy leads to greater hydrodynamic damping.

Finally, increasing the width of the float results in a higher peak value of the dimensionless excitation force, rising from approximately 0.45 for $l/H = 0.125$ to around 0.58 for $l/H = 0.375$, in the case of $H/\lambda > 0.25$. This is due to the fact that a larger floater surface intercepts a greater portion of the incident wave front, thereby receiving more energy from the wave field. The increased exposed surface enhances the diffraction effects, resulting in a stronger excitation force acting on the floater.

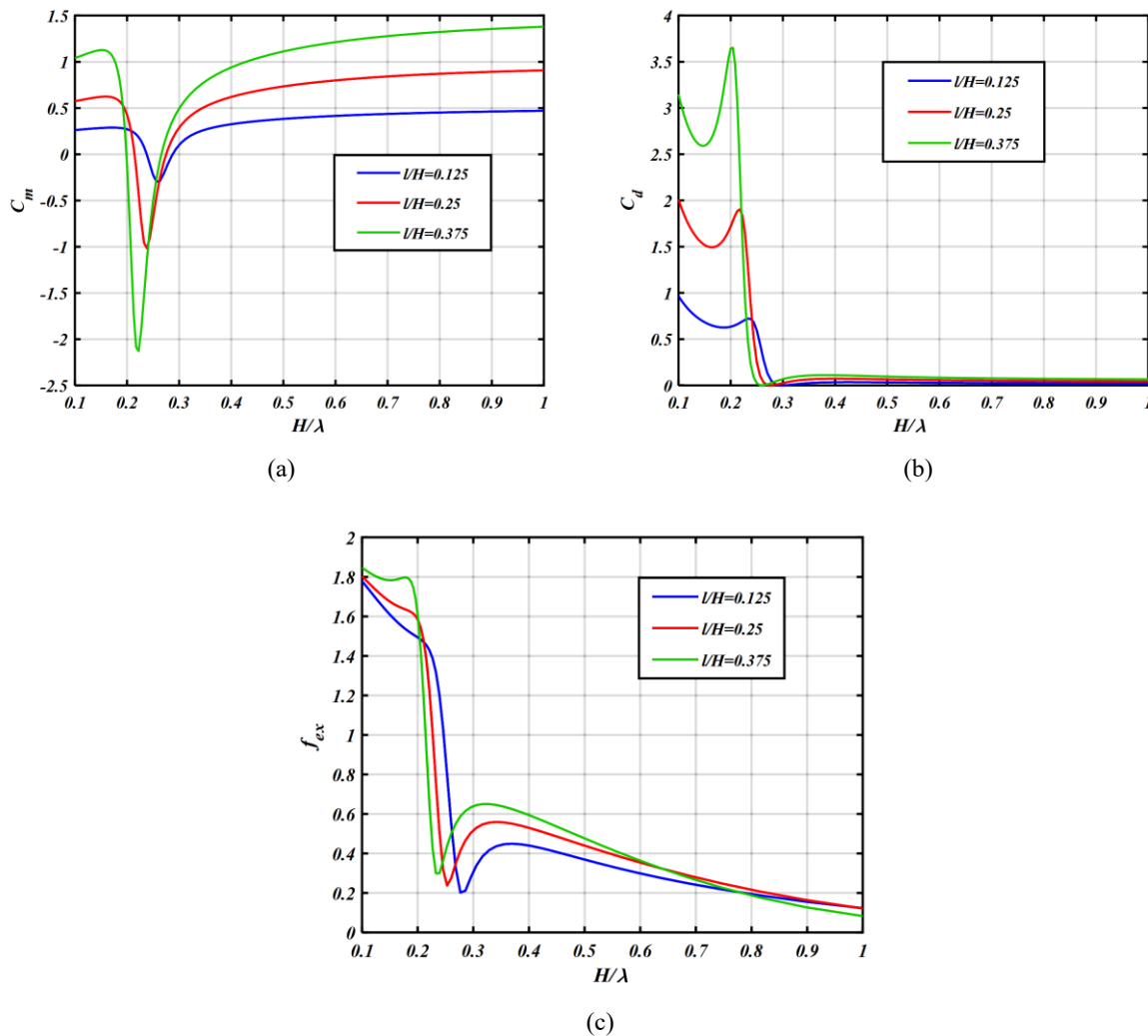


Fig. 7 Dimensionless hydrodynamic coefficients as a function of H/λ . (a): dimensionless added mass coefficient C_m (b): dimensionless damping coefficient C_d and (c): dimensionless excitation force f_{ex} .

5. Conclusion

This study provided a numerical analysis of the hydrodynamic behavior of a quayside wave energy conversion system in the presence of a rigid vertical dike. Using the Boundary Element Method (BEM), the key hydrodynamic coefficients, namely, added mass, radiation damping, and excitation force, were determined in a dimensionless form for a rectangular floating buoy operating in fluid domain of intermediate depth. The results showed good agreement with experimental and analytical data from the literature, thereby confirming the validity of the proposed model. Since the adopted BEM formulation is based on linear potential theory, the findings are primarily applicable to the low-frequency regime.

The numerical analysis highlighted the significant influence of the dike, draft depth and width of the floater on the hydrodynamic performance of the studied system. The results indicate that the proximity of the dike amplifies wave interferences, increasing radiation damping and excitation force, while greater distances reduce these effects. Moreover, increasing the draft of the floater leads to a decrease in the hydrodynamic coefficients and a shift of their peaks toward longer waves. Conversely, a wider floater increases added mass, damping, and excitation force. These findings underline the importance of geometric and environmental parameters in the design and optimization of wave energy converters integrated into existing coastal infrastructure, aiming to enhance both energy production and coastal protection.

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